

CUSTOMER SALES ANALYSIS

Project Overview

An advanced data analysis project that merges sales and customer data to identify top customers, analyze purchasing patterns, and generate business insights using Pandas and Matplotlib.

Objectives

- Merge multiple datasets (sales and customer data)
- Perform grouping and aggregation operations
- Handle datetime data for trend analysis
- Create pivot tables for data summarization
- Build visualizations to communicate insights

Setup & Installation

1. Install Python 3.x
2. Install required libraries:
`pip install pandas matplotlib jupyter`
3. Clone this repository
4. Open ``customer_analysis.ipynb`` in Jupyter/VS Code
5. Run all cells in order

Project Structure

- ``customer_analysis.ipynb`` — main analysis notebook
- ``data/`` — contains `sales_data.csv` and `customer_churn.csv`
- ``visualizations/`` — contains generated charts
- ``analysis_report.pdf`` — final report with insights

Technical Approach

1. Loaded and explored both datasets

2. Checked for missing values (none found)
3. Merged datasets using Customer_ID (handled format mismatch using regex)
4. Performed 3 types of aggregations:
 - Total sales by region
 - Average sale by contract type
 - Top 5 customers by spending
5. Created a pivot table (Region vs Contract Type)
6. Converted date column to datetime and extracted monthly trends
7. Built 4 visualizations (bar, line, pie, and churn analysis charts)

Key Findings

Revenue & Customers

- Total Revenue: ₹12,365,048
- Total Customers: 100
- Average Order Value: ₹123,650.48

Top Performers

- Top Customer: CUST016 (₹373,932 total spend)
- Top Region: North (₹3,983,635 in sales)
- Best Month: March

Business Insights

- Customers on One-Year contracts spend the most on average
- North and South regions together generate over 60% of total revenue
- Overall churn rate is 10%
- Sales peaked in March and dropped significantly in April

Charts Generated

1. Bar Chart — Total sales by region
2. Line Chart — Monthly sales trend
3. Pie Chart — Sales distribution by contract type
4. Bar Chart — Churn rate by contract type

Recommendations

1. Focus retention efforts on Month-to-month customers (higher churn risk)
2. Investigate April's sales drop for seasonal patterns
3. Replicate North region's successful strategies in West region
4. Encourage One-Year contracts through incentives (higher average spend)

What I Learned

- How to merge datasets with mismatched ID formats using regex
- Multiple types of groupby aggregations
- Creating and interpreting pivot tables
- Working with datetime data in Pandas
- Building a complete data analysis pipeline in Jupyter Notebook

Notebook Execution Summary:

The notebook was run successfully in VS Code with Jupyter extension.

All cells executed without errors, producing the following outputs:

- Data loaded (100 sales records, 500 customer records merged)
- 3 aggregations computed successfully
- Pivot table generated
- 4 visualizations created and saved

```
task5 > customer_analysis.ipynb > # =====
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables ... Python 3.14.5

# =====
# FINAL SUMMARY REPORT
# =====
print("="*50)
print("    CUSTOMER SALES ANALYSIS REPORT")
print("="*50)
print(f"Total Revenue: ₹{merged_df['Total_Sales'].sum():.2f}")
print(f"Total Customers: {merged_df['Customer_ID'].nunique()}")
print(f"Average Order Value: ₹{merged_df['Total_Sales'].mean():.2f}")
print(f"Top Customer: {top_customers.index[0]} - ₹{top_customers.iloc[0]:.2f}")
print(f"Top Region: {region_sales.index[0]}")
print(f"Best Month: {monthly_sales.idxmax()}")
print(f"Overall Churn Rate: {merged_df['Churn'].mean()*100:.2f}%")
print("="*50)

[24] ✓ 0.0s Python

... =====
    CUSTOMER SALES ANALYSIS REPORT
=====
Total Revenue: ₹12,365,048.00
Total Customers: 100
Average Order Value: ₹123,650.48
Top Customer: CUST016 - ₹373,932.00
Top Region: North
Best Month: March
Overall Churn Rate: 10.00%
=====
```

Final Analysis Report Output

Chart 1: Total Sales by Region

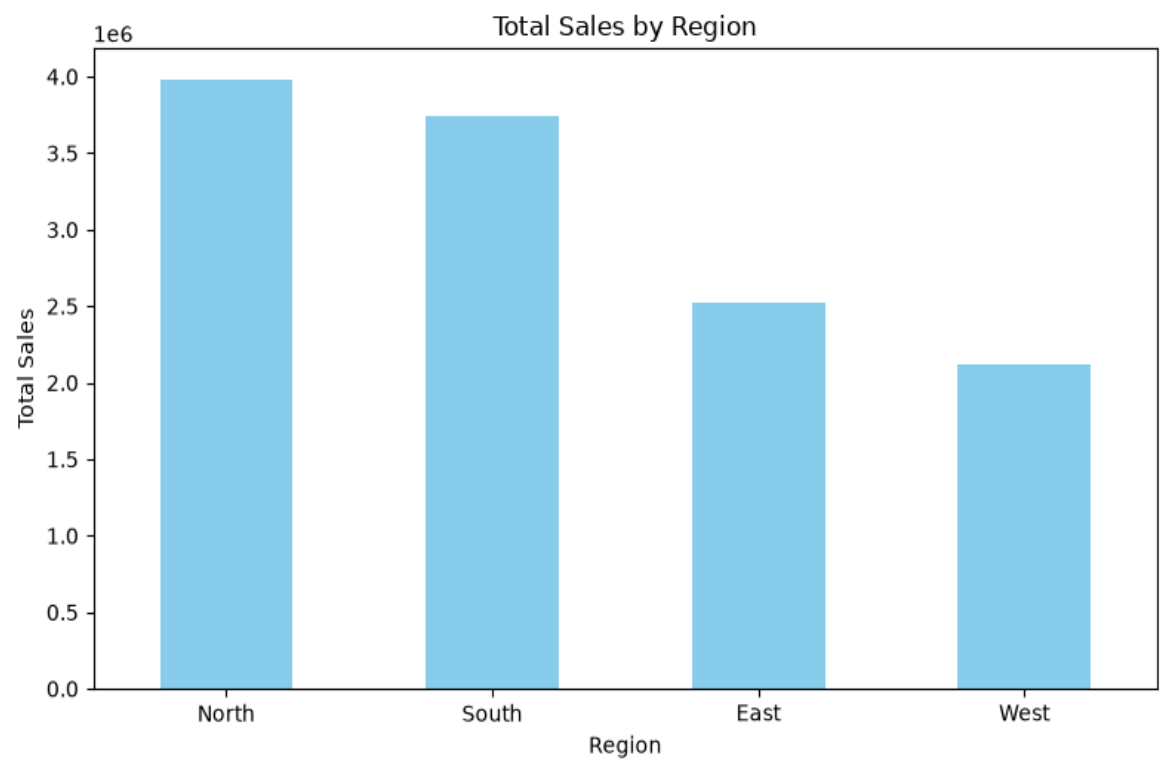


Chart 2: Monthly Sales Trend

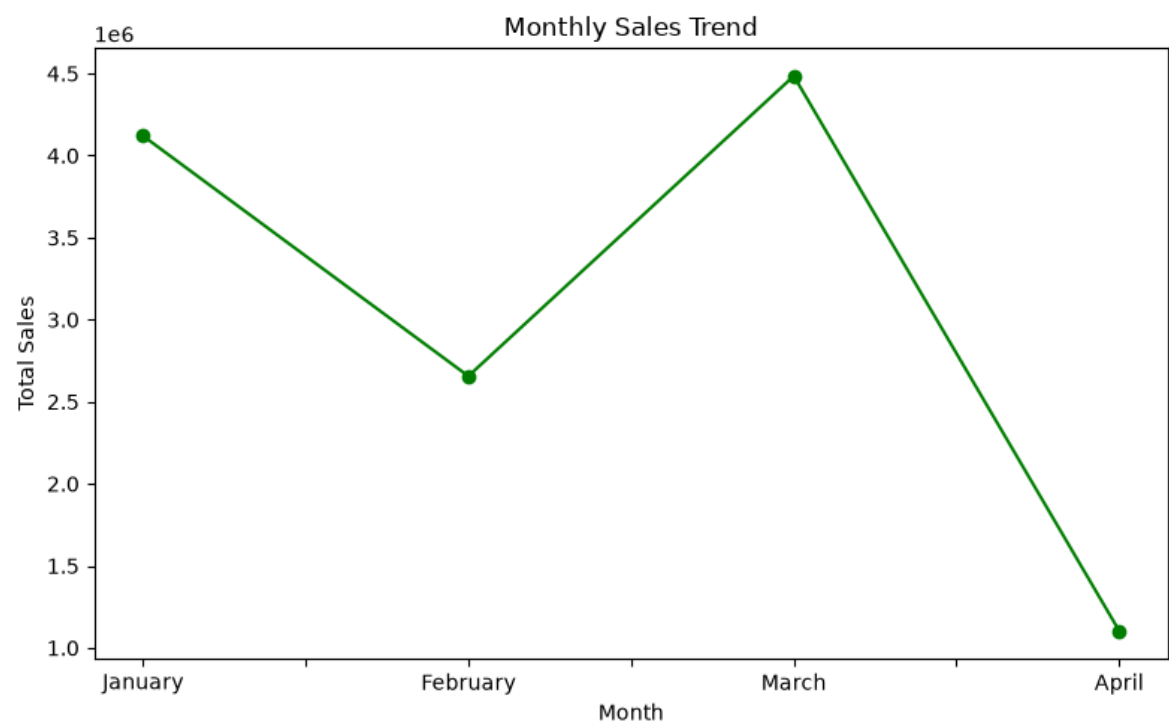


Chart 3: Sales Distribution by Contract Type

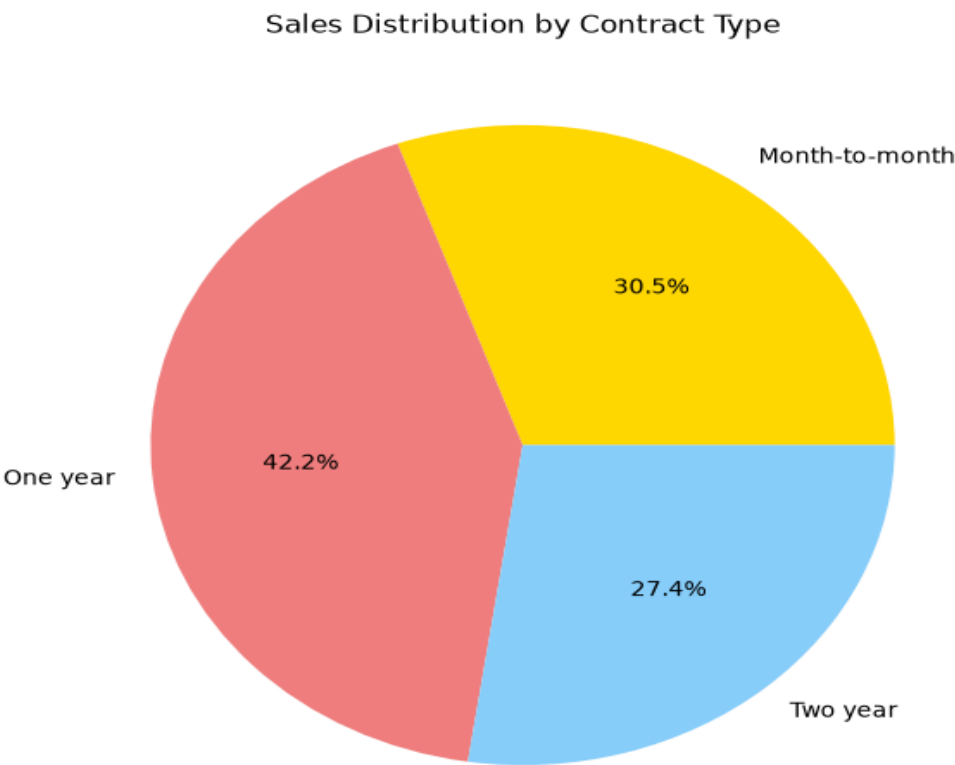


Chart 4: Churn Rate by Contract Type

